Time: 1 hr. Paper Code: CSC-501	B. Tech. (V Semester) 2nd Sessional Test 2025-26 Object Oriented Programming	Max. Marks: 12

Note: Attempt All Questions. Make suitable assumptions wherever necessary.

Q1. CO2	Implement a class *Counter* that contains a private data member *count*, constructors to initialize the value of count and a member function *get_count()* to return its current value. In addition, write the necessary code such that the following driver function runs correctly and performs the desired operations.
	
	`int main()`
	`{`
	`  Counter c1(3), c2(3);`
	`  cout << "\n c1 = " << c1.get_count();`
	`  cout << "\n c2 = " << c2.get_count();`
	`  c2 = --c1;`
	`  cout << "\n c1 = " << c1.get_count();`
	`  cout << "\n c2 = " << c2.get_count();`
	`  c2 = c1--;`
	`  cout << "\n c1 = " << c1.get_count();`
	`  cout << "\n c2 = " << c2.get_count() << endl;`
	`  return 0;`
	`}`
		[4] BL-3,4

Q2. CO2	Create a template class *SafeArray* that can store elements of any data type. The class should overload the subscript operator [ ] to perform bounds checking and display an error message if an invalid index is accessed. In the main() function, create two *SafeArray* objects of different types, such as int and float, store some values in them, and display the elements using the overloaded [ ] operator.	**[4] BL-3,4,6**
Q3. CO4	Discuss briefly the similarities and differences between abstract class and interface in Java, emphasizing their purpose, usage, and key characteristics.	**[4] BL-4**